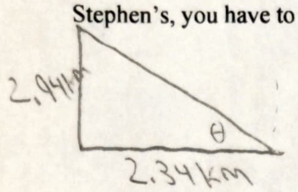
100%

PHY201 Exam 1 on chapter(s): 1, 2, 3

Name: ______________________

These questions assess your understanding of the material from Chapter(s) 1, 2, 3 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Stephen's. To get to Stephen's, you have to walk 2.34 km west and then 2.94 km north. What is the angle of a line connecting your house to his?



$\tan = \frac{o}{a}$

$\tan = \frac{2.94}{2.34}$

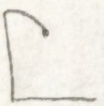
$\tan^{-1}(1.256)$

$\theta = 51.5°$ North of West

ANSWER: 51.5° North of West ✓

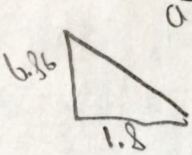
2. A BASE jumper runs off a cliff. When he loses contact with the cliff, he is traveling purely in the x-direction at a speed of 1.8 m/s. What is his overall speed after 0.7 seconds?

$V_{ox} = 1.8$ m/s
$V_{fx} = 1.8$ m/s
$V_{oy} = 0$ m/s
$V_{fy} = ?$
$a_y = 9.8$ m/s²
$t = 0.7$s



$V_f = V_o + at$
$V_f = 9.8(.7)$
$V_{fy} = 6.86$ m/s

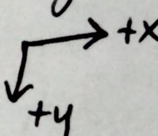
$a^2 + b^2 = c^2$



$6.86^2 + 1.8^2 =$
$V = 7.09$ m/s

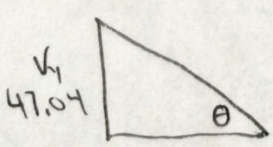
ANSWER: 7.09 m/s ✓

* Note: student is implicitly saying



3. Ian throws a water balloon horizontally from a bridge with an initial speed of 35.0 m/s. Find the angle at which the water balloon enters the water 4.8 seconds later. You may ignore air resistance.

$V_{ox} = 35$ m/s
$V_{fx} = 35$ m/s
$V_{oy} = 0$ m/s
$V_{fy} = ?$
$a_y = 9.8$ m/s²
$t = 4.8$s

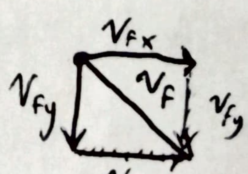


$\tan = \frac{o}{a}$
$\tan^{-1}(1.344)$
$\theta = 53.35°$

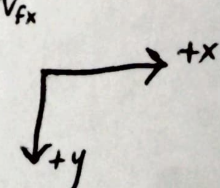
$V_f = V_o + at$
$V_f = 9.8(4.8)$
$V_{fy} = 47.04$ m/s

ANSWER: 53.35° ✓

*



Note:



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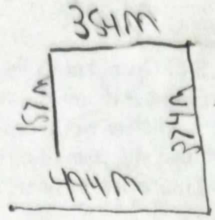
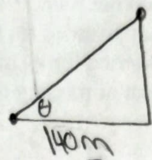
4. It takes 282 seconds for a walker to move 494 m east, then 374 m north, then 354 m west, and finally 157 m south. What is the walker's net displacement? Include the magnitude and the angle direction of the displacement vector.

$t = 282\,s$



ANSWER: 258.24m + 57.17° North of East

$a^2 + b^2 = c^2$



$140^2 + 217^2 = c^2$

$= 258.24m$

$494 + 374 + 354 + 157 = 1379m$

$\tan = \frac{o}{a} = \frac{217}{140}$

$\tan^{-1}(1.55)$

$= 57.17°$ North of East

✓

5. Jennifer is walking on a bridge between two mountain peaks, when curiosity suddenly overcomes her and she throws a laptop straight down from the bridge to the distant vale below. If the laptop falls 112 m in 2.2 s, then how far will it have fallen in 13.3 s?

$\Delta h = 112m$

$t = 2.2s$

$\Delta h = ?$

$t = 13.3s$

$a = 9.8 m/s$

(Δy)
$\Delta x = V_0 t + \frac{1}{2}at^2$

$112m = V_0(2.2) + \frac{1}{2}(9.8)(2.2)^2$

-23.7

$\frac{88.284}{2.2} = \frac{V_0(2.2)}{2.2}$

$V_0 = 40.13 m/s$

ANSWER: 1400.5 m

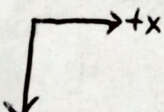
(Δy)
$\Delta x = V_0 t + \frac{1}{2}at^2$

$\Delta x = 40.13(13.3) + \frac{1}{2}(9.8)(13.3)^2$

$533.72 + 866.761$

$\Delta h = 1400.5m$

✓



6. Jeremy is driving in his car wondering how long would it take him, starting from 3.0 m/s and accelerating in a straight line at 3.7 m/s², to travel a distance of 300 m?

$V_0 = 3 m/s$

$a = 3.7 m/s^2$

$\Delta x = 300m$

$t = ?$

$V_f^2 = V_0^2 + 2a(\Delta x)$

$V_f^2 = 3^2 + 2(3.7)(300)$

$V_f^2 = 2229$

$V_f = 47.21 m/s$

ANSWER: 11.95s

$V_f = V_0 + at$

$47.21 = 3 + 3.7(t)$

-3

$t = 11.95s$

✓

+3

7. Linda boards a light-rail commuter train each day for work. One day, Linda decides to use her phone's stopwatch to measure the time it takes for the train to reach its top speed of 89.5 km/h from rest, and finds that it takes 16.0 s. What is the train's average acceleration?

$V_f = 89.5 km/h$

ANSWER: $1.55 m/s^2$

$V_0 = 0 km/h$

$t = 16s$

$\bar{a} = ?$

$\bar{a} = \frac{\Delta v}{\Delta t}$ $\quad \bar{a} = \frac{24.86}{16} = 1.55 m/s^2$

✓

$\frac{89.5km}{h} \Big| \frac{1000m}{1km} \Big| \frac{1hr}{60min} \Big| \frac{1min}{60s} = 24.86 m/s$

8. A small airplane flies in a straight line at a speed of 237 km/h. How long does it take the plane to fly 133 leagues? Report your answer in minutes. An inch is 2.54 cm, a mile is 2640 cubits, a league is 3 miles, and a cubit is 2 feet.

$V = 237 km/h = 3950 m/min$ $\quad V = \frac{\Delta x}{t}$

ANSWER: 162.56 min

$\Delta x = 133$ leagues

$3950 = \frac{642128.256}{\Delta t} = 162.56 min$

$t = ?$

✓

$\frac{133L}{} \Big| \frac{3m}{1L} \Big| \frac{2640c}{1m} \Big| \frac{2ft}{1c} \Big| \frac{12in}{1ft} \Big| \frac{2.54cm}{1in} = 642128.256m$

$\frac{237km}{1hour} \Big| \frac{1hour}{60min} \Big| \frac{1000m}{1km} = 3950 m/min$

9. A basketball referee tosses the ball straight up for the starting tip-off. What is the minimum velocity at which a basketball player must leave the ground in order to rise 3.91 m above the floor and make contact with the ball?

$V_0 = ?$

ANSWER: 8.75 m/s

$\Delta h = 3.91m$

$V_f^2 = V_0^2 + 2a(\Delta h)$

~~B~~ Student should have −9.8 here otherwise it'd be

$V_f = 0$

$0 = V_0^2 + 2(9.8)(3.91)$

$a = 9.8$

$\sqrt{76.64} = \sqrt{V_0^2}$

$V_0 = \sqrt{-76.64}$ which is impossible for real numbers

$V_0 = 8.75 m/s$

✓

+3

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1. 38.5° west of north

2. 7.1 m/s

3. 53.3° above the horizontal

4. 258 m, 57.2° north of east

5. 1400 m

6. 11.9 s

7. 1.55 m/s^2

8. 163 minutes

9. 8.75 m/s